
Professional Summary

Computer Engineering undergraduate with professional experience in software performance engineering and machine learning research. Experienced in microservice architecture, automated testing frameworks, and cloud-based deployments using Python and Docker. Interested in the intersection of AI-enhanced systems and engineering leadership to deliver performant, intelligent software solutions.

Education

University of Toronto, Toronto, Ontario, CA September 2021 - May 2026 Expected
Bachelor of Applied Science in Computer Engineering (CE) Candidate

- AI Engineering Minor, Engineering Leadership Certificate
- Incoming Master of Engineering (MEng) Candidate, University of Toronto (Starting Fall 2026)

Professional Experience

Intern Software Performance Engineer, Veeva Systems – Toronto, CA May 2024 – August 2025

- Developed automated testing frameworks (Python, JMeter, Shell), reducing performance analysis time.
- Diagnose performance bottlenecks to improve the reliability of the cloud applications.
- Collaborated with developers and QAs across teams in an Agile environment (Scrum/Kanban) to streamline Jenkins pipeline performance testing.

Research Assistant, Dynamic Optimization & Ops Mgmt Lab – Toronto, CA May 2023 – August 2023

- Developed baseline ML models with Scikit-Learn to predict taxi-in/out times and runway assignments.
- Applied data preprocessing (NumPy, Seaborn) and exploratory analysis to improve prediction accuracy.
- Contributed to project progress through literature review, team collaboration, and problem-solving.

Selected Engineering Projects

[VIVIDpdf](#) – **Text-to-Speech Web Application** (Capstone Project) September 2025 – In Progress

- Developed an AI-driven TTS JavaScript web app for reading out loud PDF documents.
- Developed features: selective playback, local reading history, custom pronunciation and skipping rules.
- Deployed on Cloudflare to allow ease of access without the need to download an app.

[PixelNova](#) – **Productivity Management Web App** (Course Project) September 2025 – December 2025

- Developed a Flask & PostgreSQL microservice using JWT for session management and user authentication.
- Architected inter-service communication via an API Gateway, ensuring data flow across the microservices.
- Containerized and deployed the service using Docker and Render, managing migrations and environment configurations.

Web Search Engine Web Application (Course Project) September 2025 – October 2025

- Developing a Python-based search engine implementing a web crawler, inverted index, and query processor.
- Designed document and Word ID mappings for efficient indexing and retrieval.
- Configured and managed AWS EC2 instances to host and test web components.

Autonomous Route-Following Car (Hackathon Project)

December 2023

- Developed an autonomous robotic car by using microcontroller components, motor drivers, and sensor arrays.
- Implemented control algorithms to detect course boundaries and navigate a predefined route.
- Optimized system reliability under time constraints, demonstrating problem-solving and rapid prototyping.

Smart Distance-Based Music Player (Hackathon Project)

February 2023

- Built an Arduino UNO and electronics prototype to detect moving objects and play context-specific music.
- C++-programmed microcontroller logic for distance measurements and dynamic sound output.
- Collaborated in a 4-person team, integrating hardware and software within a 24-hour hackathon timeframe.

Leadership & Community Involvement

- **Professional Mentorship:** Onboarded and mentored a new team member, facilitating their transition into team workflows, daily operations, and organizational culture.
- **Academic Collaboration:** Spearheaded peer study groups and curated shared digital resource repositories to enhance collaborative learning among classmates.
- **Student Advocacy:** Led a successful initiative to modernize the University of Toronto's student identity verification process for new admissions.
- **International Student Support:** Provided cultural and academic orientation for incoming students, assisting with their transition to the Canadian educational environment.
- **Community Youth Club (HK):** Participate in fundraising campaigns and community service projects, including elder care visits and environmental conservation efforts.

Skills & Interests

Technical:

- **Frameworks/Tools:** Flask, React, JMeter, Scikit-Learn, NumPy, Seaborn, Jenkins, Git
- **Cloud & DevOps:** AWS (EC2), Docker, PostgreSQL, Render, CI/CD Pipeline optimization
- **Specializations:** Software Performance Engineering, Microservices, AI/Machine Learning, Embedded Systems (Arduino/Microcontrollers)

Languages:

- English, Chinese, Japanese (Beginner)
- Python, C/C++, Java, JavaScript, SQL, Shell Scripting, HTML, CSS